

Azure Cloud Fundamentals and Data Pipeline

Implementation using ADF

Task 1: Explore Azure Portal & Create Resource Group

Objective

To understand the Azure Portal and create a Resource Group for organizing Azure resources

Procedure

1. Logged into the Microsoft Azure Portal using the Azure for Students account.
2. Searched for **Resource Groups** in the Azure Portal.
3. Clicked on **Create**.
4. Selected the Azure for Students subscription.
5. Entered the Resource Group name as **RG-ADF-Assignment**.
6. Selected the appropriate region.
7. Clicked **Review + Create** and then **Create**.
8. The Resource Group was created successfully.

Result

The Resource Group **RG-ADF-Assignment** was successfully created in Microsoft Azure under the **Azure for Students** subscription and will be used to manage all resources required for this assignment.

Figure1: Resource Group Created Successfully

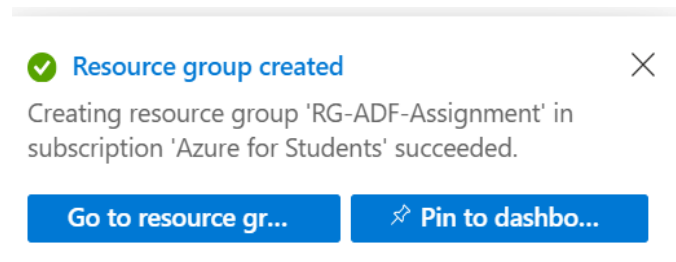


Figure2: Resource Group Listed in Azure Portal

☐	Name ↑	Subscription	Location
☐	 RG-ADF-Assignment	⋮ Azure for Students	Central India

Task 2: Storage Setup

Objective

To create an Azure Storage Account, create a Blob Container, and upload the Superstore CSV dataset.

Procedure

1. Logged into the Microsoft Azure Portal using the Azure for Students subscription.
2. Created an Azure Storage Account named **ssuperstore22** under the Resource Group **RG-ADF-Assignment**.

3. Selected **Standard** performance and **Locally Redundant Storage (LRS)** redundancy.
4. Opened the Storage Account and navigated to **Containers**.
5. Created a Blob Container named **superstore-data**.
6. Created a **source** directory inside the Blob Container.
7. Uploaded the **superstore_raw.csv** dataset to the **source** directory.

Result

The Azure Storage Account and Blob Container were successfully created in Microsoft Azure. The **superstore_raw.csv** dataset was successfully uploaded to the source directory in Azure Blob Storage and is ready to be processed using Azure Data Factory.

Figure 3: Storage Account Created Successfully (*Deployment Succeeded popup*)

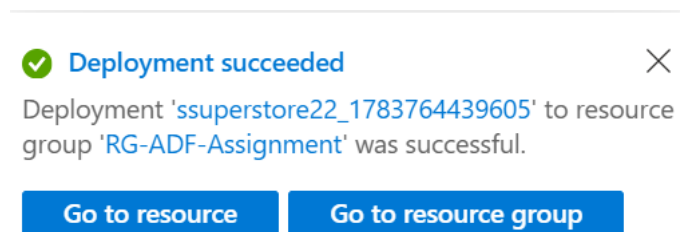


Figure 4: Storage Account Overview

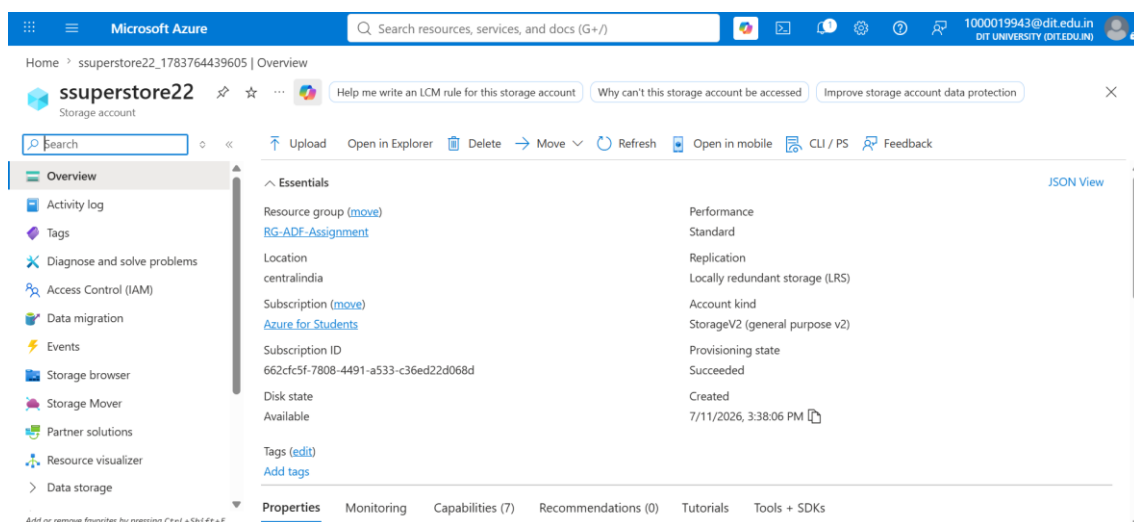


Figure 5: Blob Container Created Successfully

✓

Successfully created storage container

✕

Successfully created storage container 'superstore-data'.

Figure 6: Blob Container (superstore-data)

Home > ssuperstore22_1783764439605 | Overview > ssuperstore22

ssuperstore22 | Containers

Storage account

Search

+

Add container

↑

Upload

↺

Refresh

🗑

Delete

🔒

Change access level

↶

Restore containers

⌵

Edit columns

Search containers by prefix

Only show active containers

Showing all 2 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector	
☐	📁 \$logs	7/11/2026, 3:38:29 PM	Private	Available	-	⋮
☐	📁 superstore-data	7/11/2026, 3:42:19 PM	Private	Available	-	⋮

Add or remove favorites by pressing Ctrl+Shift+F

Figure 7: superstore_raw.csv Uploaded Successfully to the Source Directory

superstore-data

Container

Search

+

Add Directory

↑

Upload

↺

Refresh

🗑

Delete

📄

Copy

📄

Paste

🔄

Rename

🔑

Acquire lease

🔑

Break lease

⌵

Edit columns

superstore-data > source

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	📁 [-]						⋮
☐	📄 superstore_raw.csv	7/11/2026, 3:48:26 PM	Hot (Inferred)	Block blob	2.18 MiB	Available	⋮

Task 3: Azure Data Factory (ADF Basics)

Objective

To create an Azure Data Factory, connect it with Azure Blob Storage using a Linked Service, create source and destination datasets, and use the Get Metadata activity to retrieve information about the uploaded CSV file.

Procedure

1. Logged into the Microsoft Azure Portal.
2. Navigated to **Data Factories**.
3. Created a new Azure Data Factory named **adf-sneha22** under the Resource Group **RG-ADF-Assignment**.
4. Selected **Central India** as the region.
5. Successfully deployed the Azure Data Factory.
6. Opened **Azure Data Factory Studio** by clicking **Launch Studio**.
7. Explored the **Author**, **Monitor**, and **Manage** sections available in Azure Data Factory Studio.
8. Opened the **Manage** section in Azure Data Factory Studio.
9. Created a **Linked Service** named **LS_BlobStorage**.
10. Connected the Linked Service to the Azure Blob Storage account (**ssuperstore22**) using the Azure for Students subscription.
11. Verified the connection and successfully created the Linked Service.
12. Created a source dataset named **DS_Source** using the **DelimitedText** format
13. Connected the dataset with the **LS_BlobStorage** linked service
14. Configured the dataset to read the superstore_raw.csv file stored in the source directory of the superstore-data Blob container

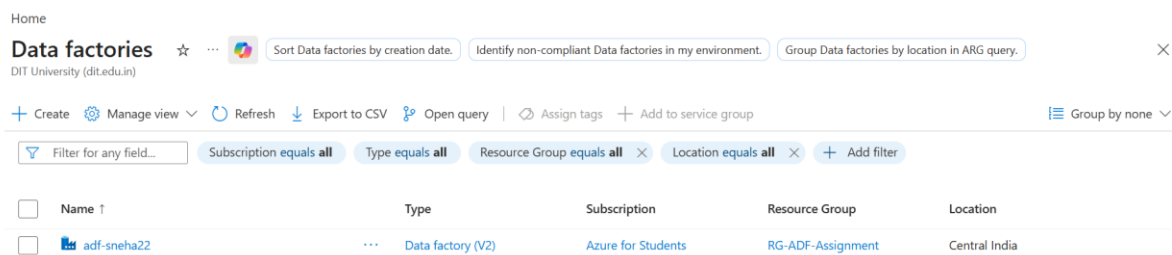
15. Published the dataset to Azure Data Factory.
16. Created a destination dataset named **DS_Destination** using the **DelimitedText** format.
17. Connected the dataset with **LS_BlobStorage**.
18. Configured the dataset to write the output file **superstore_output.csv** into the **destination** folder of the **superstore-data** Blob container.
19. Published the dataset to Azure Data Factory.
20. Created a pipeline named **PL_Superstore** in Azure Data Factory.
21. Added a **Get Metadata** activity to the pipeline.
22. Configured the **Get Metadata** activity to use the **DS_Source** dataset for retrieving file metadata.
23. Added a **Copy Data** activity to the pipeline.
24. Configured the **Source** of the Copy Data activity by selecting the **DS_Source** dataset.
25. Configured the **Sink** of the Copy Data activity by selecting the **DS_Destination** dataset.
26. Connected the **Get Metadata** activity to the **Copy Data** activity using the **Success** dependency.
27. Validated the pipeline to ensure there were no configuration errors.
28. Published all changes to Azure Data Factory.
29. Executed the pipeline using the **Debug** option.
30. Verified that both the **Get Metadata** and **Copy Data** activities completed successfully.

31. Confirmed that the output file **superstore_output.csv** was successfully created in the **superstore-data** Azure Blob Storage container.

Result

The Azure Data Factory pipeline was successfully created and executed. The Linked Service was connected to Azure Blob Storage, and both the source and destination datasets were configured successfully. The **Get Metadata** activity retrieved metadata from the source file, and the **Copy Data** activity copied the data from **superstore_raw.csv** to **superstore_output.csv**. The successful execution was verified in Azure Data Factory, and the output file was confirmed in the Azure Blob Storage container.

Figure 8: Azure Data Factory Created Successfully



The screenshot shows the Azure Data Factory portal interface. At the top, there's a 'Home' link and a 'Data factories' section with a star icon and a dropdown menu. Below this, there are several filter buttons: 'Subscription equals all', 'Type equals all', 'Resource Group equals all', and 'Location equals all'. A table lists the data factories, with one entry visible: 'adf-sneha22'.

Name	Type	Subscription	Resource Group	Location
adf-sneha22	Data factory (V2)	Azure for Students	RG-ADF-Assignment	Central India

Figure 9: Azure Data Factory Studio Interface

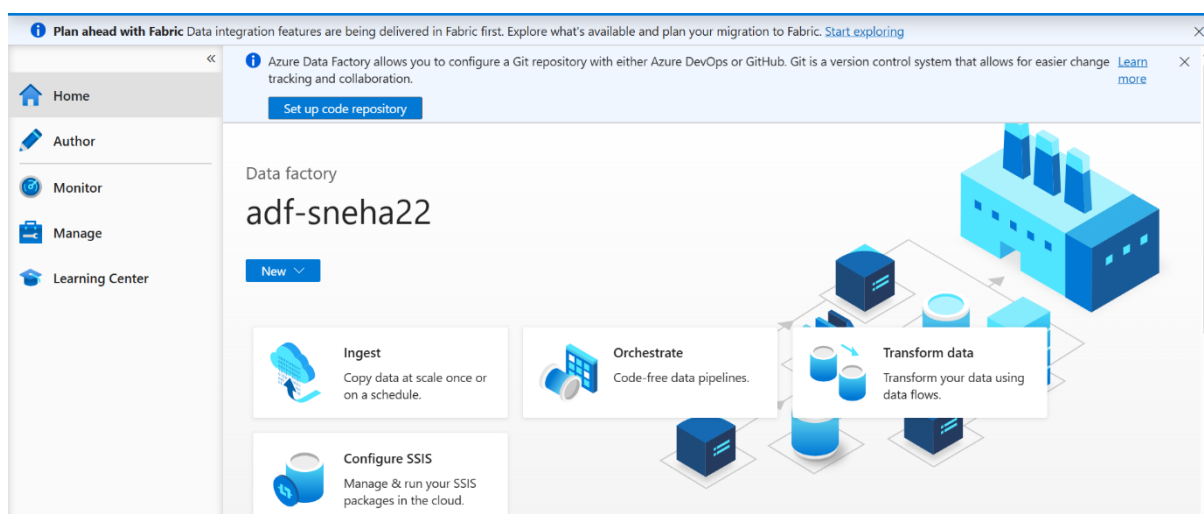


Figure 10: Source Dataset (DS_Source) Configuration in Azure Data Factory

The screenshot shows the configuration for a Source Dataset named **DS_Source** in Azure Data Factory. The interface includes a left-hand navigation pane with **Factory Resources** (Pipelines: 0, Change Data Capture (preview): 0, Datasets: 1, DS_Source selected, Data flows: 0, Power Query: 0). The main area displays the **DS_Source** configuration with a **DelimitedText** icon and a **CSV** file icon. The **Connection** tab is active, showing the following settings:

- Linked service ***: LS_BlobStorage
- File path ***: superstore-data / source / supe
- Compression type**: No compression
- Column delimiter**: Comma (,)
- Row delimiter**: Default (\r\n, or \n)

Buttons for **Test connection**, **Edit**, **+ New**, and **Learn more** are visible. The **Properties** pane on the right shows the **General** tab with fields for **Name *** (DS_Source), **Description**, and **Annotations** (+ New). A **Cancel** button is at the bottom right.

Figure 11: Destination Dataset (DS_Destination) Configuration

The screenshot shows the configuration for a Destination Dataset named **DS_Destination** in Azure Data Factory. The interface includes a left-hand navigation pane with **Factory Resources** (Pipelines: 0, Change Data Capture (preview): 0, Datasets: 2, DS_Destination selected, DS_Source, Data flows: 0, Power Query: 0). The main area displays the **DS_Destination** configuration with a **DelimitedText** icon and a **CSV** file icon. The **Connection** tab is active, showing the following settings:

- Linked service ***: LS_BlobStorage
- File path ***: superstore-data / destination / supe
- Compression type**: No compression
- Column delimiter**: Comma (,)
- Row delimiter**: Default (\r\n, or \n)

Buttons for **Test connection**, **Edit**, **+ New**, and **Learn more** are visible. The **Properties** pane on the right shows the **General** tab with fields for **Name *** (DS_Destination), **Description**, and **Annotations** (+ New). A **Cancel** button is at the bottom right.

Figure 12 – Pipeline Design

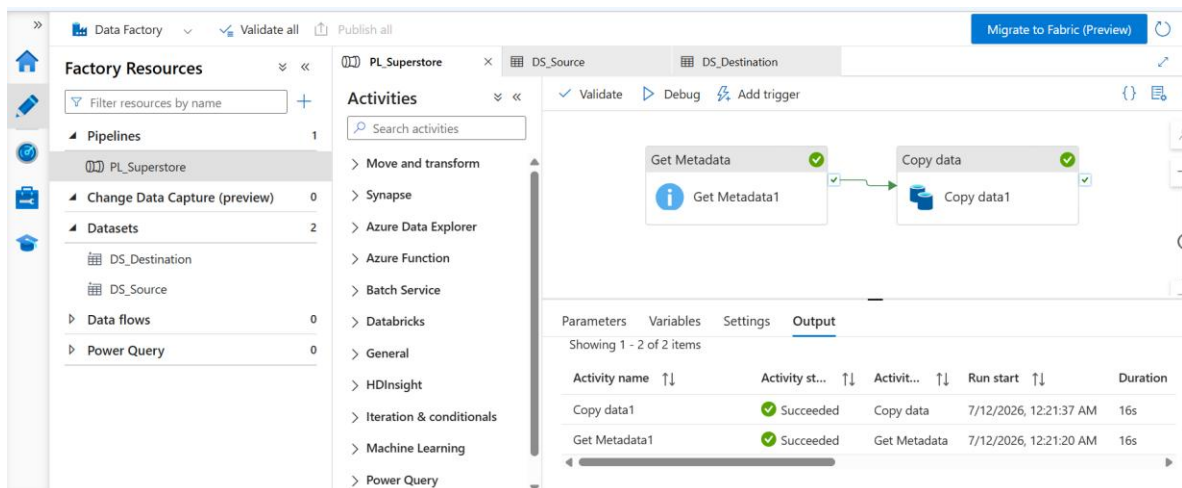
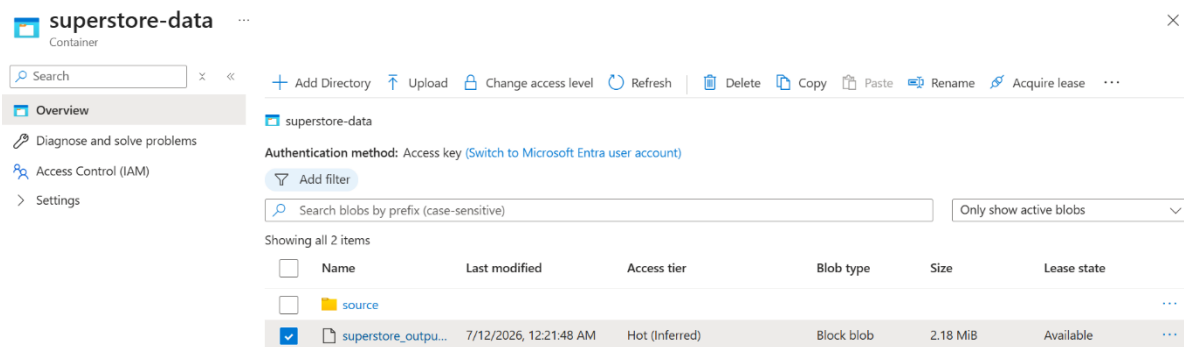


Figure 13 – Output File (superstore_output.csv) Successfully Created in Azure Blob Storage



Task 4: Pipeline Development

Objective

To develop an Azure Data Factory pipeline that copies data from a source CSV file stored in Azure Blob Storage to a destination location using the **Copy Data** activity.

Procedure

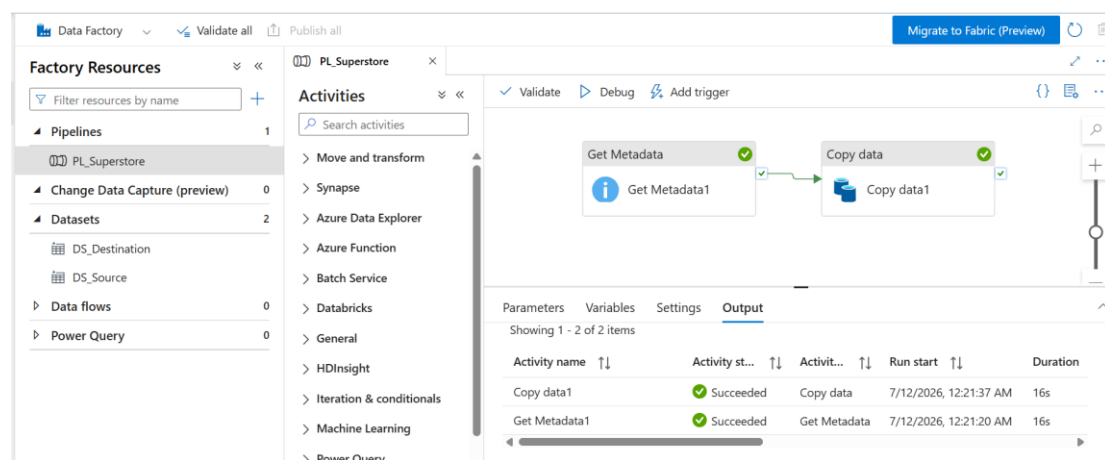
1. Opened **Azure Data Factory Studio** and navigated to the **Author** section.
2. Created a new pipeline named **PL_Superstore**.

3. Added the **Get Metadata** activity to validate the source file before processing.
4. Added the **Copy Data** activity to the pipeline.
5. Configured the **Source** by selecting the **DS_Source** dataset.
6. Configured the **Sink** by selecting the **DS_Destination** dataset.
7. Connected the **Get Metadata** activity to the **Copy Data** activity using the **Success** dependency.
8. Validated the pipeline to ensure there were no configuration errors.
9. Published all changes to Azure Data Factory.
10. Executed the pipeline using the **Debug** option.
11. Verified that the data was successfully copied from **superstore_raw.csv** to **superstore_output.csv** in Azure Blob Storage.

Result

The **PL_Superstore** pipeline was successfully developed using the **Copy Data** activity. The source and destination datasets were configured successfully, and the pipeline copied the data from the source CSV file to the destination file in Azure Blob Storage. The successful execution of the pipeline was verified in Azure Data Factory.

Figure 14: Pipeline Design



Task 5: Pipeline Execution

Objective

To execute the Azure Data Factory pipeline using the Debug option and verify that all pipeline activities complete successfully.

Procedure

1. Opened Azure Data Factory Studio.
2. Navigated to the Author section and opened the PL_Superstore pipeline.
3. Clicked the Debug button to execute the pipeline.
4. Waited for the pipeline execution to complete.
5. Monitored the execution status in the Output tab.
6. Verified that both the Get Metadata and Copy Data activities completed with the Succeeded status.
7. Confirmed that the output file superstore_output.csv was successfully created in Azure Blob Storage.

Result

The **PL_Superstore** pipeline executed successfully using the **Debug** option. Both the **Get Metadata** and **Copy Data** activities completed with the **Succeeded** status. The successful execution confirmed that the data was copied from **superstore_raw.csv** to **superstore_output.csv** in Azure Blob Storage without any errors.

Figure 15: Successful Pipeline Execution

Parameters

Variables

Settings

Output

Showing 1 - 2 of 2 items

Activity name	↑↓	Activity st...	↑↓	Activit...	↑↓	Run start	↑↓	Duration
Copy data1		✔ Succeeded		Copy data		7/12/2026, 12:21:37 AM		16s
Get Metadata1		✔ Succeeded		Get Metadata		7/12/2026, 12:21:20 AM		16s

Task 6: IAM Roles

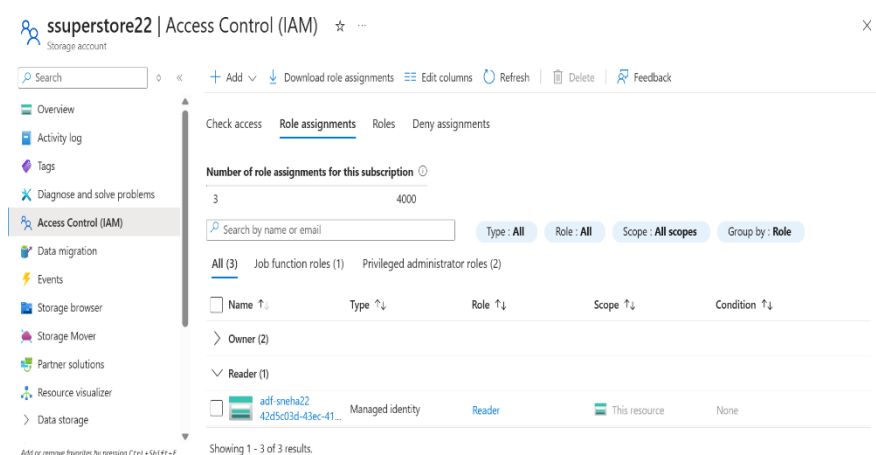
Procedure

1. Opened the Azure Portal and navigated to the Azure Storage Account (ssuperstore22).
2. Opened the Access Control (IAM) section.
3. Clicked Add → Add role assignment.
4. Selected the Reader role.
5. Assigned the Reader role to the Azure Data Factory managed identity (adf-sneha22).
6. Verified that the role assignment was successfully completed.
7. Attempted to assign the Contributor role; however, the role was not available due to permission restrictions in the Azure for Students subscription.
8. Confirmed that the Azure Data Factory managed identity had been granted the available IAM access.

Result

The **Reader** role was successfully assigned to the Azure Data Factory managed identity (**adf-sneha22**) for the Azure Storage Account (**ssuperstore22**). The **Contributor** role was not available for assignment because of Azure subscription permission restrictions.

Figure 16: Reader Role Assignment to Azure Data Factory Managed Identity



Mini Project: Azure Data Factory Pipeline for CSV Data Processing

Problem Statement

Build a complete pipeline that reads a CSV file from Azure Blob Storage and processes it using Azure Data Factory.

Objective

The objective of this mini project is to create an end-to-end Azure Data Factory pipeline that reads a CSV file from Azure Blob Storage, validates its metadata, copies the data to a new destination, and verifies successful execution.

Azure Services Used

- Azure Data Factory
- Azure Blob Storage
- Linked Service
- Source Dataset
- Destination Dataset
- Get Metadata Activity
- Copy Data Activity

Procedure

1. Created an Azure Data Factory named adf-sneha22.
2. Created a Linked Service (LS_BlobStorage) to connect Azure Data Factory with Azure Blob Storage.
3. Created a source dataset (DS_Source) for the superstore_raw.csv file located in the source folder of the superstore-data container.

4. Created a destination dataset (DS_Destination) to store the output file superstore_output.csv in the destination location.
5. Created a pipeline named PL_Superstore.
6. Added a Get Metadata activity to validate the availability and metadata of the source CSV file.
7. Added a Copy Data activity to copy the CSV data from the source dataset to the destination dataset.
8. Connected the Get Metadata activity with the Copy Data activity.
9. Validated and published the pipeline.
10. Executed the pipeline using the Debug option.
11. Verified that both Get Metadata and Copy Data activities completed successfully.
12. Confirmed that the superstore_output.csv file was successfully created in Azure Blob Storage.

Expected Output

- Source CSV file successfully detected.
- Metadata successfully validated.
- Pipeline executed successfully.
- Data copied to the destination location.
- Output CSV file created in Azure Blob Storage.

Result

The Azure Data Factory pipeline was successfully developed and executed. The pipeline validated the source file metadata using the Get Metadata activity and copied the data from the source CSV file to the destination CSV file using the Copy Data activity. The output file superstore_output.csv was

successfully generated in Azure Blob Storage, confirming successful data integration.

Figure 17: Azure Data Factory Pipeline Design (Get Metadata → Copy Data)

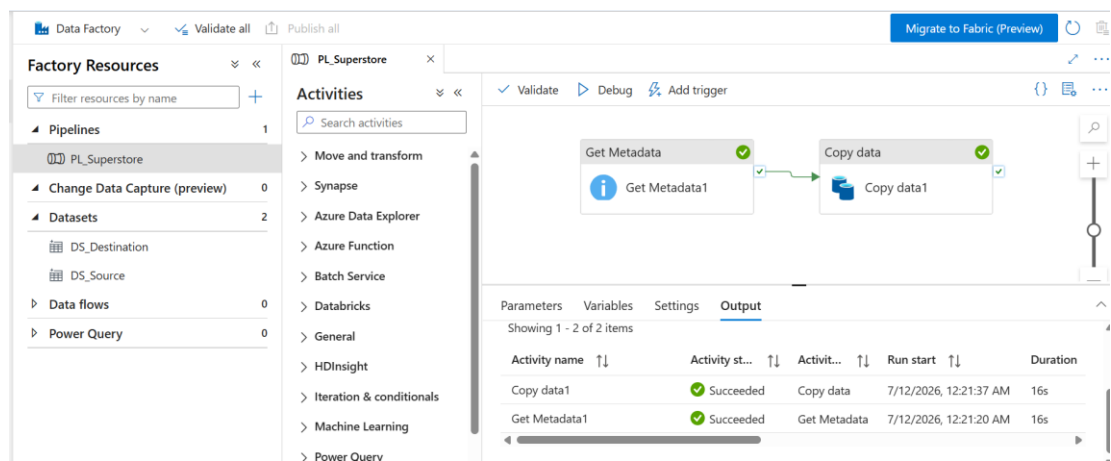


Figure 18: Successful Pipeline Execution (Succeeded)

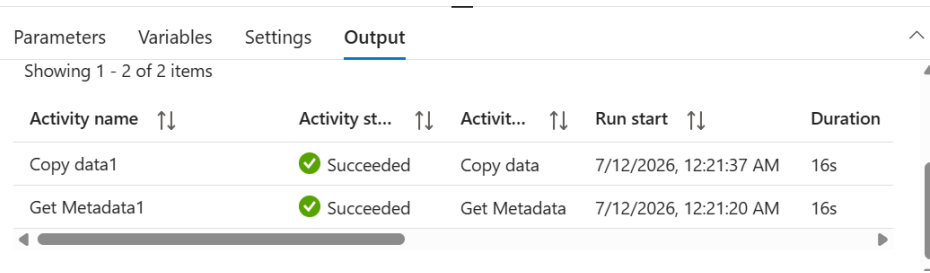
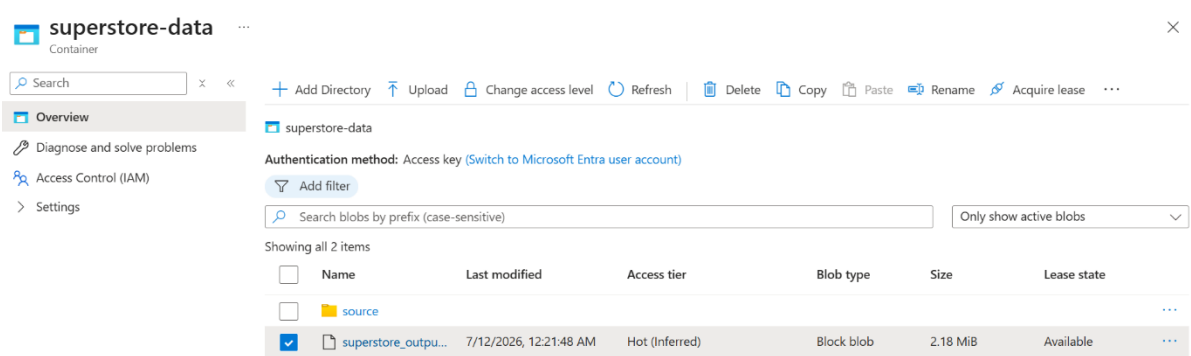


Figure 19: Output File (superstore_output.csv) in Azure Blob Storage



Conclusion

This project successfully demonstrated the use of Azure Data Factory (ADF) for building an end-to-end data integration pipeline. A Linked Service was created to connect Azure Data Factory with Azure Blob Storage, and both source and destination datasets were configured. The pipeline was developed using Get Metadata and Copy Data activities to validate the source file and copy data to the destination location. The pipeline executed successfully, and the output file was generated in Azure Blob Storage. This project provided practical experience in creating, managing, and executing data pipelines using Azure cloud services.

Learning Outcomes

- Understood the Azure Data Factory interface (Author, Monitor, and Manage).
- Created and configured Linked Services and Datasets.
- Built and executed data pipelines using Get Metadata and Copy Data activities.
- Connected Azure Data Factory with Azure Blob Storage.
- Verified successful pipeline execution and data movement.
- Gained practical experience with cloud-based ETL and data integration.

